



VayuSetu

Find the pollution hotspot on your street,
prove it, and get it fixed.

Citizen photos, ground stations and satellites, verified by Gemini and routed to the agency that can clear it.

12

cities live, India + BRICS

184

ground stations live

13

languages

4

Gemini jobs, all grounded

Live: cleanair-backend--cleanair-clear-streets.asia-southeast1.hosted.app

THE PROBLEM

A city-wide AQI hides the air people breathe

99%

of people live where WHO air-quality guidelines are not met

4.2 M

premature deaths from outdoor air pollution in 2019

89%

of those deaths are in low- and middle-income countries



Street-level sources go unseen

A garbage fire or a construction site kilometres from the nearest monitor vanishes into the city average.



Pollution crosses borders

Crop burning and haze drift across state and national lines faster than one agency can track.



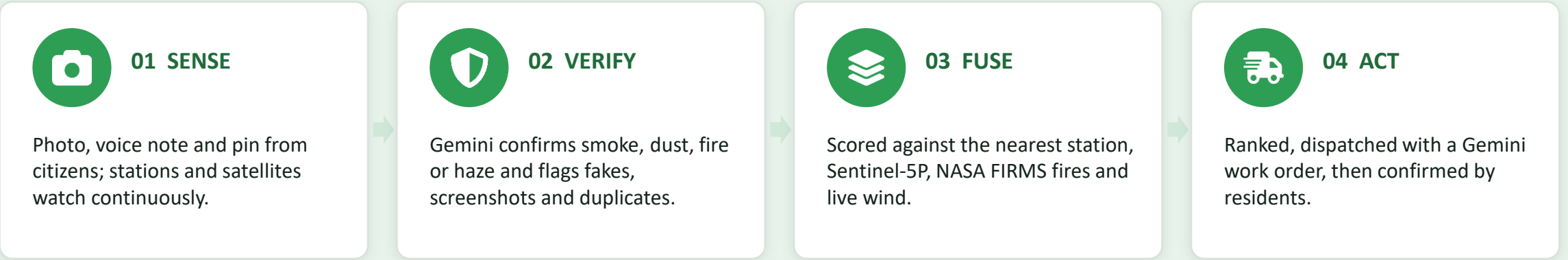
Data rarely reaches a crew

Dashboards publish numbers, but nobody is told which street to inspect or which department owns the fix.



One monitor, many invisible sources, and smoke that does not stop at the border.

From a citizen's photo to a verified, dispatched fix

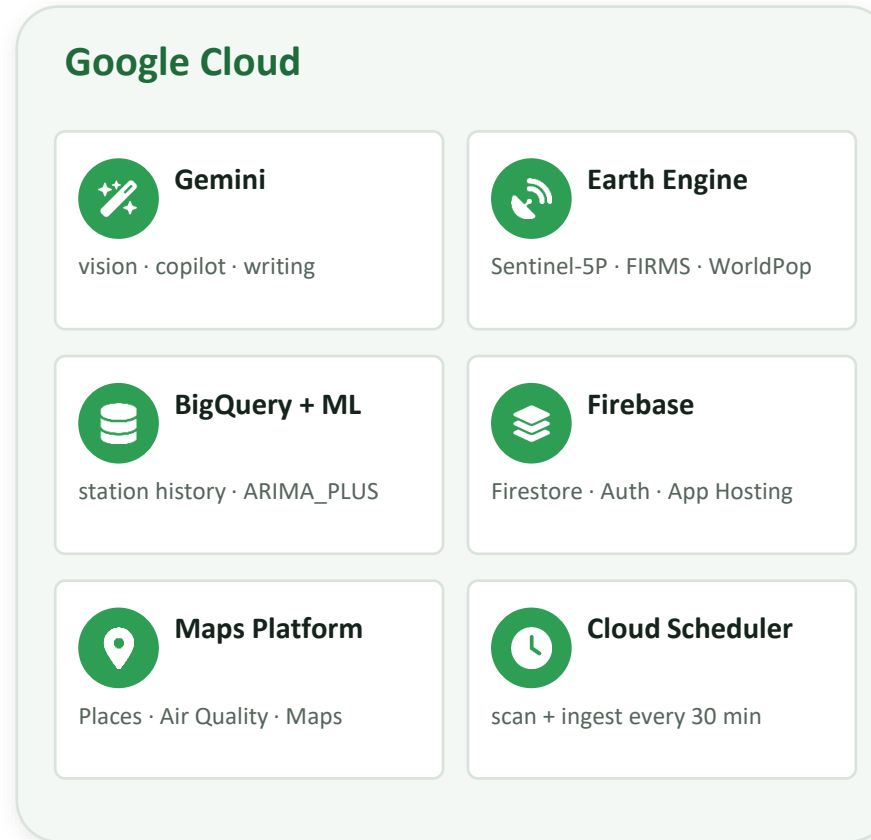
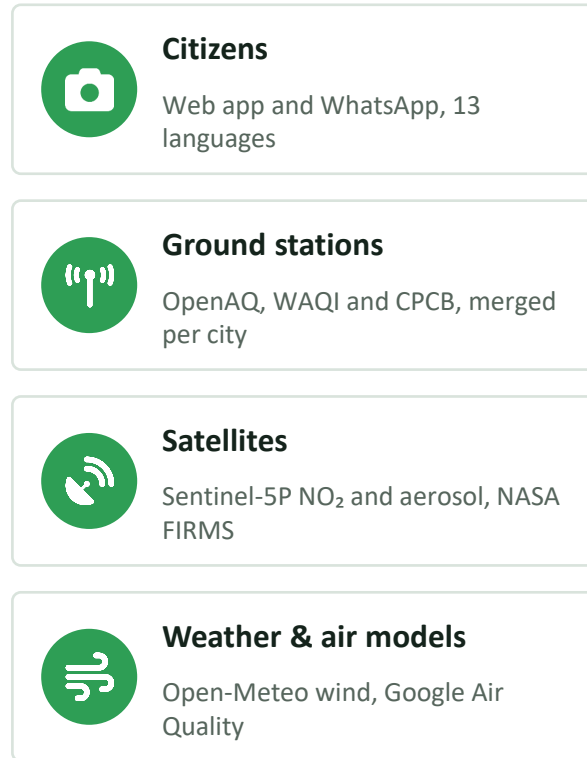


Not another AQI dashboard

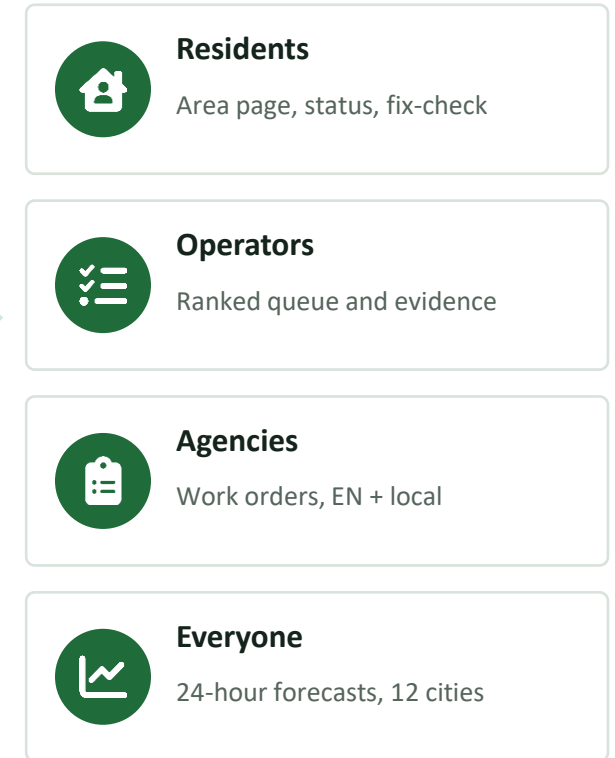
	TYPICAL AQI DASHBOARD	VAYUSETU
Scale	One city-wide average	Street-level H3 cells of ~0.7 km ²
Evidence	A single sensor feed	Citizens + stations + Sentinel-5P + FIRMS + wind
Trust	Reports taken at face value	Gemini, EXIF and duplicate checks; independent agreement
Outcome	Displays a number	Ranked queue, routed work order, residents confirm the fix

One hexagon grid ties every source together

INPUTS



OUTPUTS



Key decision: every report, station reading and satellite pixel is indexed to the same H3 hexagon ($\sim 0.7 \text{ km}^2$), so different evidence can be compared in the same place.

Gemini does four jobs and shows its evidence



Photo check

Labels smoke, dust, fire or haze with severity, and flags screenshots, edits and stock images.

Vision + JSON output



Operations copilot

Answers operators by calling live tools: incidents, stations, wind, fires, forecasts, upwind sources.

Function calling · 7 tools



Work orders

Drafts the order for the right agency, citing the evidence, in English and the city's language. For a repeat hotspot, asks for action at the source.

Structured output



Resident summaries

Explains what's happening on each street in the reader's language, with read-aloud.

13 languages · cached

Guardrails

Facts in, facts cited

Prompts carry only recorded evidence; outputs list what they used.

Never blank, never faked

Flash-Lite fallback; if both fail, the page says so.

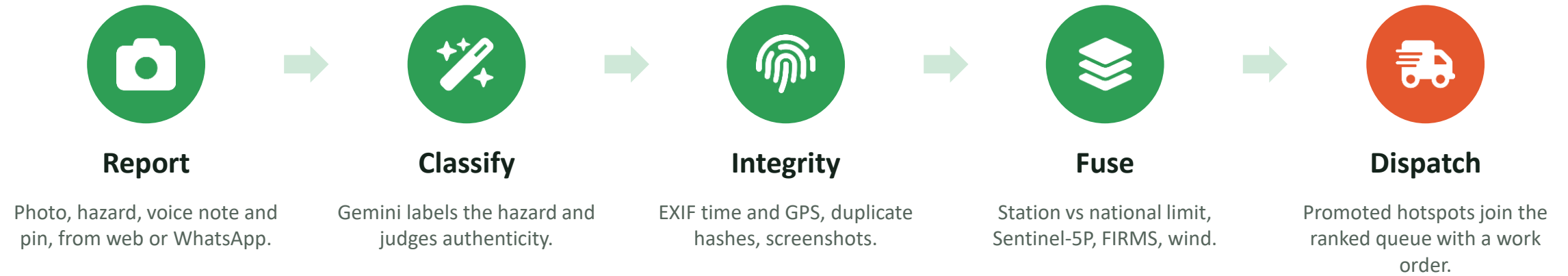
Cached per cell

Summaries regenerate only when the evidence changes.

Health advice

Comes from Google's Air Quality API, not from Gemini.

No alert without independent evidence

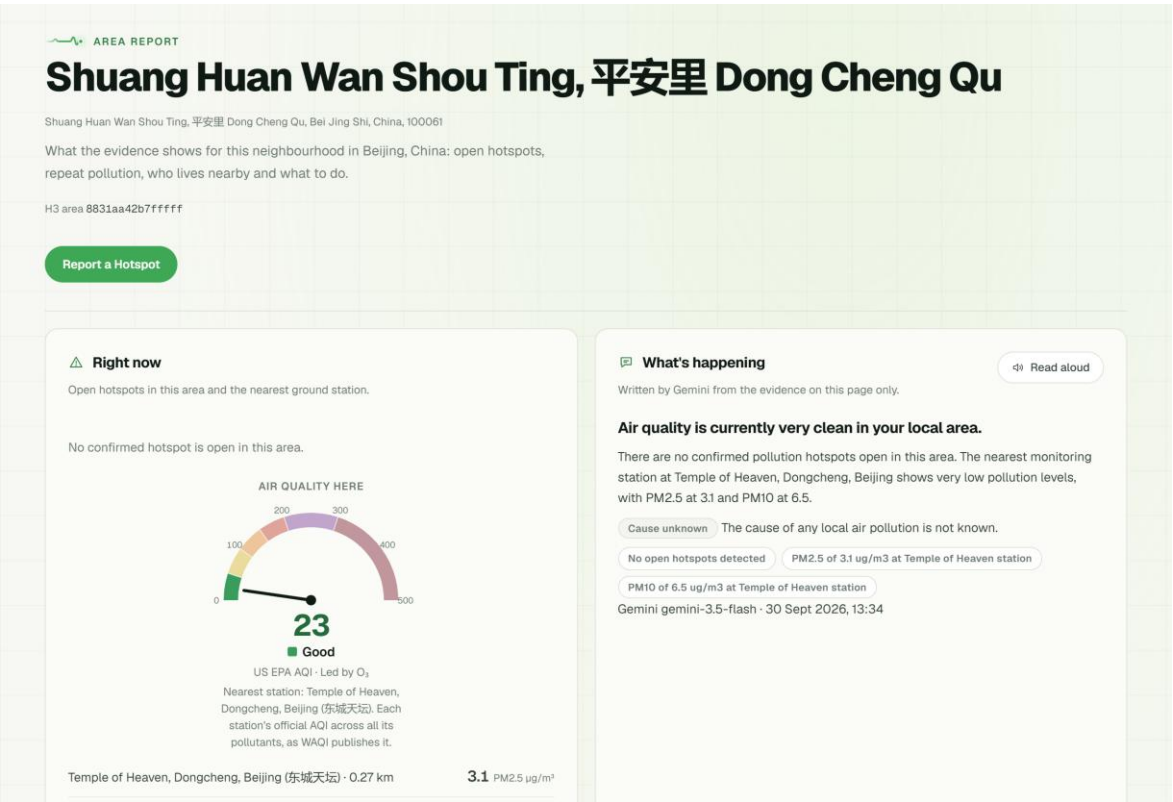
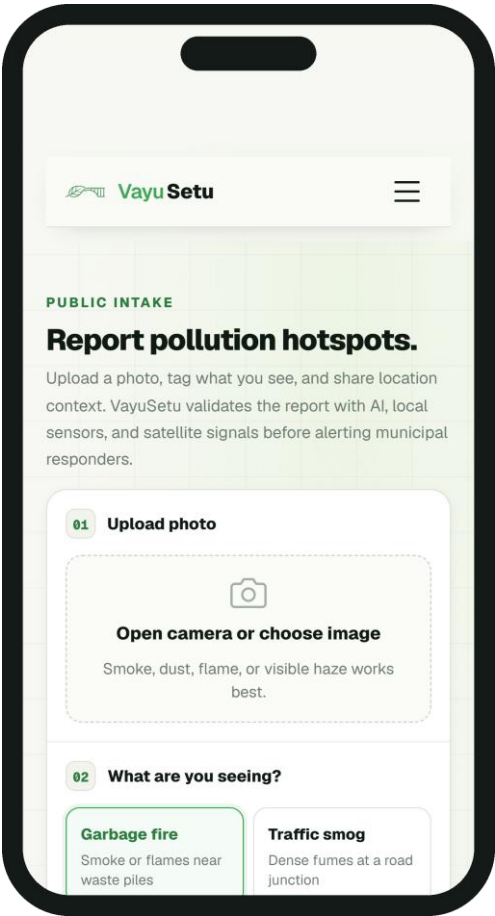


Promotion tiers, strongest first



57 of 59 incidents on record were raised automatically: a scan checks every station and Sentinel-5P every 30 minutes, with no photo needed. Flagged reports stay visible but never count toward promotion.

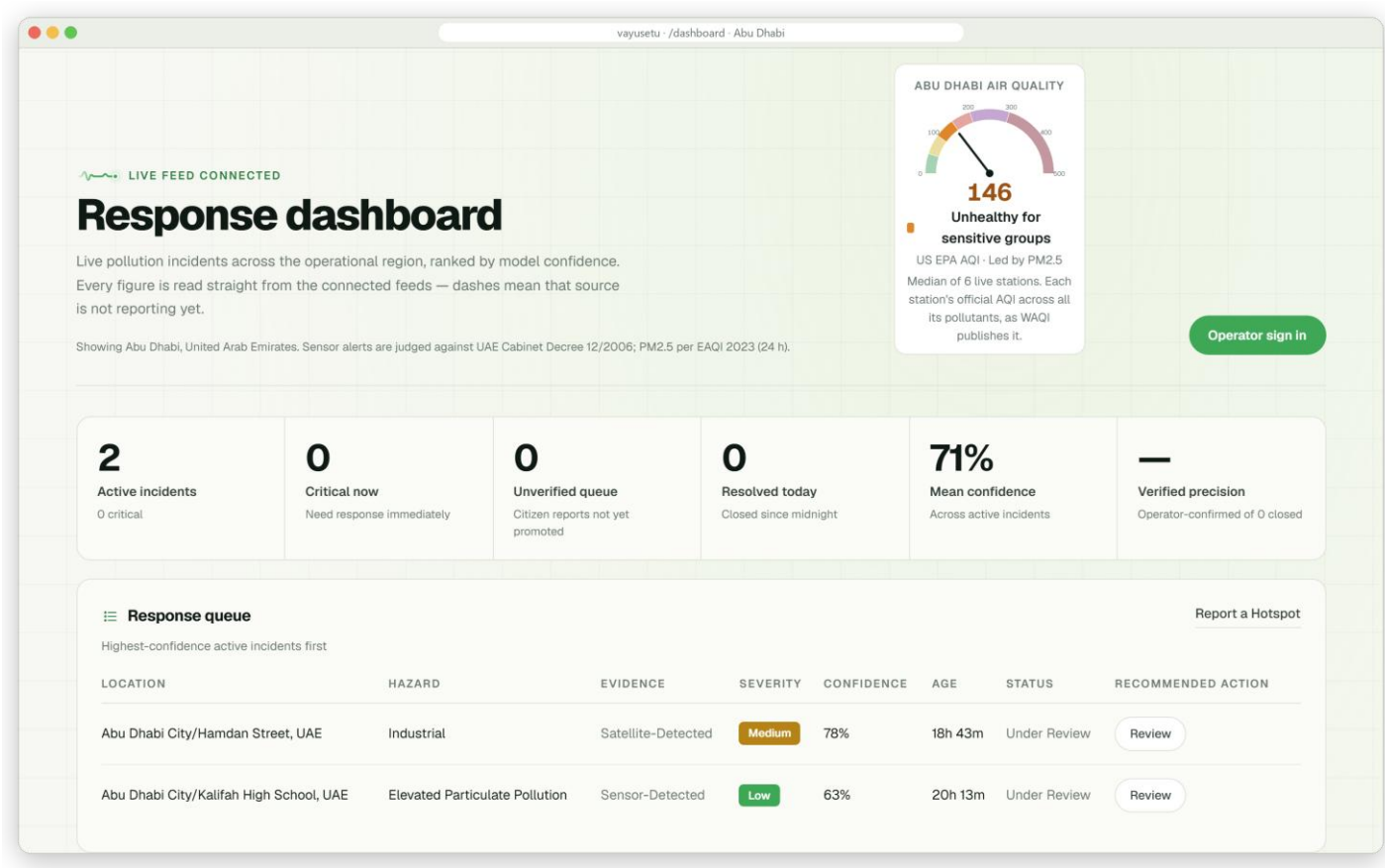
Report in seconds, see what's on your street



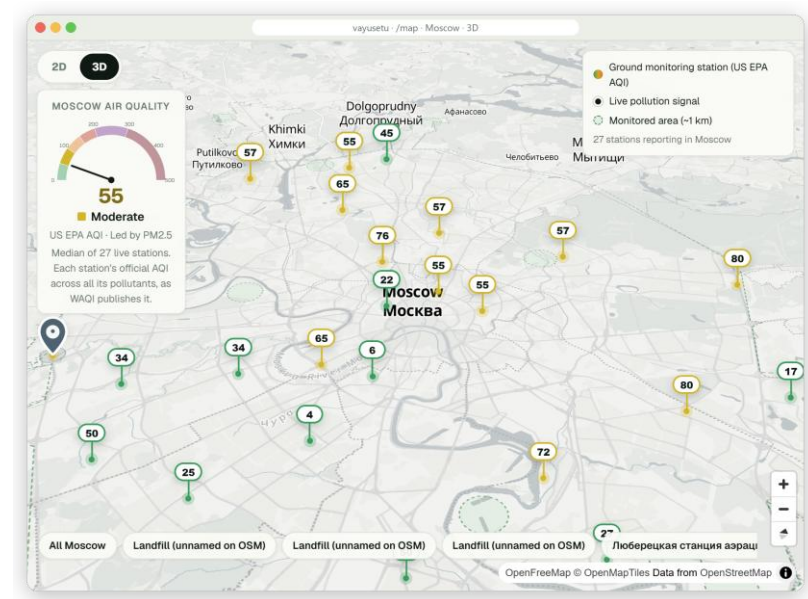
Live, 30 Sep: Temple of Heaven, Beijing. AQI 23 (US EPA, led by O₃) from a station 0.27 km away; ≈71,838 residents within 1 km.

-  **Photo, voice, pin**
13 languages; Gemini checks the photo first.
-  **A page for every street**
AQI gauge, station, Gemini summary, history.
-  **Who's affected**
WorldPop residents, schools and hospitals within 1 km.
-  **"Is it fixed?"**
Reporters confirm or dispute a fix for 72 h.

Operators see what to fix first, and why



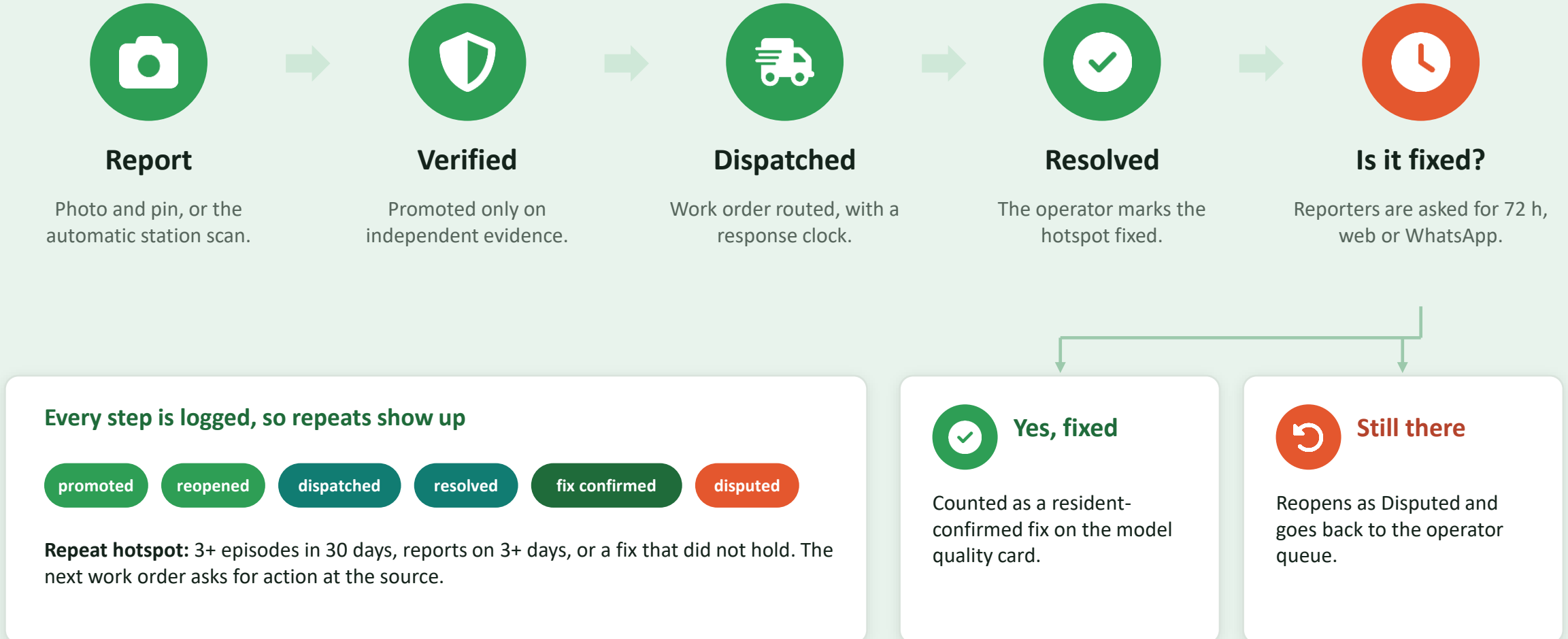
Abu Dhabi, live: AQI 146, unhealthy for sensitive groups; two hotspots raised automatically from official EAD stations, judged against the UAE limit.



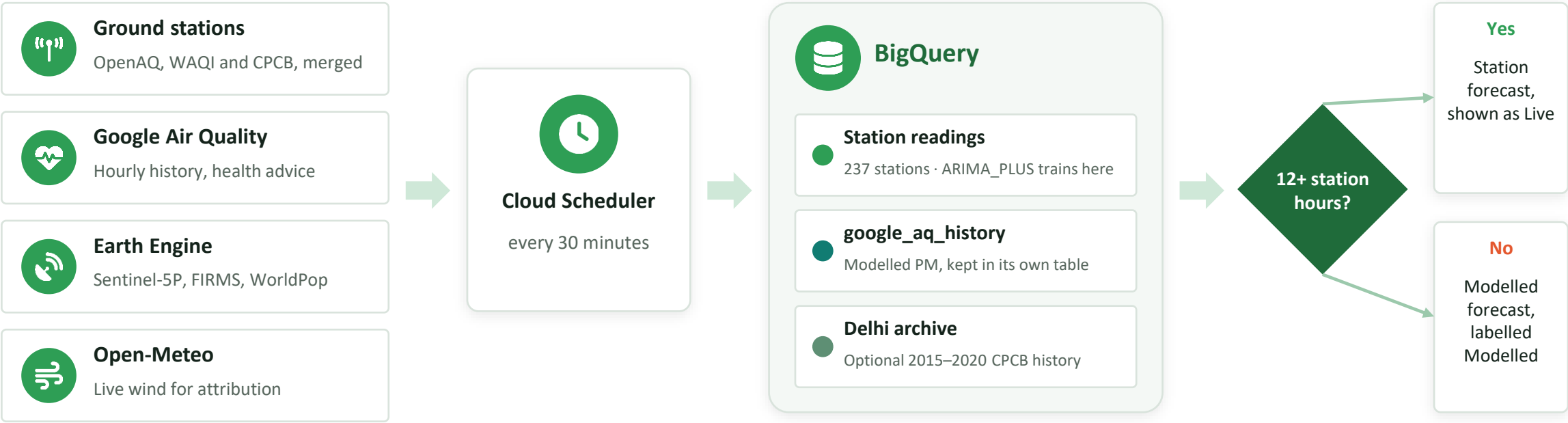
Moscow in 3D: every pin is a station's own AQI. City median 55, Moderate.

- Ranked queue**
Evidence tier first; repeat, overdue and disputed chips.
- Traceable evidence**
Station vs national limit, satellite window, reports.
- One-click action**
Draft a work order, dispatch, resolve; reporters are told.

A fix counts only when residents confirm it



From raw feeds to an honest 24-hour outlook



12 / 12

cities with a 24-hour forecast

184

ground stations live, feeding BigQuery

52

forecast zones with a week of modelled history

17,334

modelled hourly rows stored

Real station readings always win; modelled history is Google's 500 m estimate and never appears as live. Counts as of 30 Sep 2026.

Twelve cities, seven legal limits, one engine



184 ground stations live across 12 cities

57 of 59 hotspots raised automatically

≈71,838 residents counted near one Beijing landmark

**Indian cities**

60 $\mu\text{g}/\text{m}^3$
PM2.5 24 h, all six

**Beijing**

75 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

**Moscow**

35 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

**Pretoria**

40 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

**Abu Dhabi**

60 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

**Jakarta**

55 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

**Brasília**

50 $\mu\text{g}/\text{m}^3$
PM2.5 24 h limit

WHAT'S NEXT

Clean air, street by street, across India and BRICS.

Next 90 days

01

Pilot with one municipal team

Real dispatches and their response targets, instead of our placeholders.

02

Alerts near you

A WhatsApp message when a hotspot is confirmed within 1 km.

03

Proof of the fix

An after-photo from the crew, checked by Gemini against the report.

Built on Google Cloud

Gemini

Earth Engine

BigQuery ML

Firebase

Maps Platform

Air Quality API

Cloud Scheduler

Citizens

report in their language and see what happened

Operators

act on ranked, evidence-backed hotspots

Nations

share one engine and keep their own rules

Thank you

